

Concursul de matematică ”Ștefan Musta”

Ediția XXX, 5 aprilie 2025

Clasa a IX-a

Barem de corectare

Problema 1

Rezolvare: a) $2(x - [x]) + [x] = \frac{1}{2},$ (1p)

$$[x] = \frac{4x-1}{2}$$

$$[x] \stackrel{\text{not}}{\cong} k, k \in \mathbb{Z} \Rightarrow k = \frac{4x-1}{2} \Rightarrow x = \frac{2k+1}{4}, k \in \mathbb{Z}$$

Dar $[x] \leq x \leq [x] + 1$ deci $k \leq \frac{2k+1}{4} < k+1 \Rightarrow$ (1p)

$$\left. -\frac{3}{2} < k \leq \frac{1}{2} \right\} \Rightarrow k \in \{-1; 0\} \Rightarrow x = -\frac{1}{4} \text{ și } x = \frac{1}{4}. \quad (1p)$$

b) $(ab + bc + ca) \left| 1 - \frac{1}{abc} \right| = |ab + bc + ca| \cdot \left| 1 - \frac{1}{abc} \right| = \left| (ab + bc + ca) \cdot \left(1 - \frac{1}{abc} \right) \right| =$ (1p)

$$= \left| ab - \frac{1}{c} + bc - \frac{1}{a} + ca - \frac{1}{b} \right| \leq \left| ab - \frac{1}{c} \right| + \left| bc - \frac{1}{a} \right| + \left| ca - \frac{1}{b} \right|. \quad (1p)$$

c) $(|a| + |b| + |c|) \left(\frac{1}{|a|} + \frac{1}{|b|} + \frac{1}{|c|} \right) = 1 + \frac{|a|}{|b|} + \frac{|a|}{|c|} + \frac{|b|}{|a|} + 1 + \frac{|b|}{|c|} + \frac{|c|}{|a|} + \frac{|c|}{|b|} + 1 =$ (1p)

$$= 3 + \underbrace{\frac{|a|}{|b|} + \frac{|b|}{|a|}}_{\geq 2} + \underbrace{\frac{|a|}{|c|} + \frac{|c|}{|a|}}_{\geq 2} + \underbrace{\frac{|b|}{|c|} + \frac{|c|}{|b|}}_{\geq 2} \geq 3 + 2 + 2 + 2 = 9. \quad (1p)$$

Problema 2

Rezolvare: a) Fie A mulțimea numerelor divizibile cu 3 și B mulțimea numerelor divizibile cu 7

$$\text{card}(A \cup B) = \text{card } A + \text{card } B - \text{card}(A \cap B) =$$
 (1p)

$$= \left\lfloor \frac{100}{3} \right\rfloor + \left\lfloor \frac{100}{7} \right\rfloor - \left\lfloor \frac{100}{21} \right\rfloor = 33 + 14 - 4 = 43. \quad (1p)$$

$$\text{card } C(A \cup B) = 100 - 43 = 57. \quad (1p)$$

b) $\Rightarrow a_1 + 2a_2 + 2a_3 + \dots + 2a_{n-1} + a_n = 2S_n - (a_1 + a_n) =$
 $= 2 \frac{(a_1 + a_n) \cdot n}{2} - (a_1 + a_n) = (n-1)(a_1 + a_n). \quad (2p)$

$$\Leftrightarrow a_1 + 2a_2 + 2a_3 + \dots + 2a_{n-1} + a_n = (n-1)(a_1 + a_n) \Leftrightarrow$$

$$2a_1 + 2a_2 + 2a_3 + \dots + 2a_{n-1} + 2a_n = n(a_1 + a_n) \Leftrightarrow$$

$$a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n = \frac{(a_1 + a_n) \cdot n}{2} = S_n.$$

Concluzia..... (2p)

Problema 3

Rezolvare: Avem $\overrightarrow{AM} = k\overrightarrow{MB}$; $\overrightarrow{BN} = k\overrightarrow{NC}$; $\overrightarrow{CP} = k\overrightarrow{PA}$ (2p)

Dacă G este centrul de greutate al $\Delta ABC \Rightarrow$

$$\overrightarrow{GM} = \frac{\overrightarrow{GA} + k\overrightarrow{GB}}{1+k}; \overrightarrow{GN} = \frac{\overrightarrow{GB} + k\overrightarrow{GC}}{1+k}; \overrightarrow{GP} = \frac{\overrightarrow{GC} + k\overrightarrow{GA}}{1+k}; \quad (2p)$$

$$\overrightarrow{GM} + \overrightarrow{GN} + \overrightarrow{GP} = \dots = \overrightarrow{GA} + \overrightarrow{GB} + \overrightarrow{GC} = \vec{0} \Rightarrow \quad (2p)$$

G este centrul de greutate al ΔMNP . (1p)